

Student Instructions:

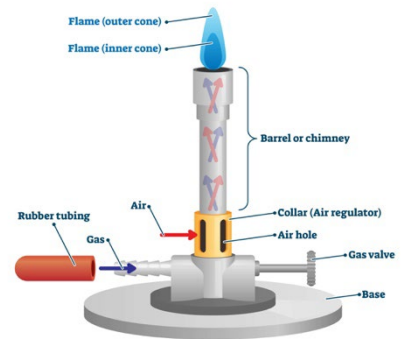
- Write your complete ID code on your answer sheet.
 - If you change your answer, erase well.
1. Which one of the following is the LEAST dense?
 - A. a lead pipe
 - B. a brick
 - C. nitrogen
 - D. water
 2. Examples include cement, wood, and metal. Which adjective below means you can't see through an object because light can't pass through it?
 - A. crystalline
 - B. opaque
 - C. translucent
 - D. transparent
 3. Which of the following choices is a chemical change?
 - A. burning wood
 - B. crushing a can
 - C. breaking glass
 - D. boiling water
 4. Below are four simple machines. An axe is an example of which one, used for lifting, tightening, or splitting?
 - A. pulley
 - B. screw
 - C. lever
 - D. wedge
 5. Gravity is one of the four fundamental ones. Which of the following physics words, measured in newtons, is the push or pull on an object?
 - A. energy
 - B. speed
 - C. velocity
 - D. force
 6. What part of the electromagnetic spectrum produced the medical image seen here?
 - A. microwaves
 - B. x-rays
 - C. infrared radiation
 - D. ultraviolet rays
 7. Which one of the metals below is NOT magnetic?
 - A. iron
 - B. steel
 - C. aluminum
 - D. nickel



8. At over 1,800 miles thick, it makes up about 84% of Earth's volume. Which choice below names this middle layer made mostly of solid rock?
- mantle
 - inner core
 - crust
 - outer core
9. Some scientists call it a fossil resin. Which of the following substances is made from tree sap and can preserve the bodies of insects?
- petrified wood
 - loam
 - sand
 - amber
10. Match each rock with its category.
- | | |
|------------|----------------|
| x. shale | 1. metamorphic |
| y. marble | 2. igneous |
| z. granite | 3. sedimentary |
- x-3, y-1, z-2
 - x-2, y-1, z-3
 - x-1, y-2, z-3
 - x-3, y-2, z-1
11. Below are four steps of the water cycle. In which one does gravity help water move through soil's layers?
- precipitation
 - evaporation
 - condensation
 - percolation
12. The vernal equinox, one of the two days each year with nearly equal amounts of daylight and darkness, occurs during which season in the Northern Hemisphere?
- fall
 - winter
 - spring
 - summer
13. In 2006, which of the following celestial bodies did the IAU downgrade from a regular planet to a dwarf planet?
- Pluto
 - Venus
 - Mars
 - Mercury
14. Two are named Crab and Horsehead. Which choice below refers to an interstellar cloud of gas and dust?
- asteroid
 - nebula
 - meteor
 - comet

15. Which of the following choices is the watery, gel-like substance that contains a eukaryotic cell's organelles, microtubules, and filaments?
- cell wall
 - nucleus
 - cell membrane
 - cytoplasm
16. Some archaea are thermophiles, meaning they live in environments with high what?
- temperatures
 - altitude
 - acid levels
 - salt levels
17. Which organism below is a type of fungus?
- an amoeba
 - a euglena
 - a mushroom
 - a paramecium
18. Three of the following trees are deciduous. Which one is an evergreen?
- dogwood
 - fir
 - hickory
 - birch
19. Which choice below names the vascular tissue that transports food from the leaves to the rest of a plant?
- sepal
 - phloem
 - cambium
 - xylem
20. Match each invertebrate animal with its phylum.
- | | |
|---------------|---------------|
| x. sea sponge | 1. Arthropoda |
| y. tarantula | 2. Porifera |
| z. oyster | 3. Mollusca |
- x-1, y-2, z-3
 - x-2, y-3, z-1
 - x-2, y-1, z-3
 - x-1, y-3, z-2
21. Platyhelminthes, Nematoda, and Annelida are respectively “flat,” “round,” and “segmented” phyla of what invertebrates?
- bees
 - crabs
 - octopi
 - worms

22. Which one of the following vertebrates has skeletal parts both externally and internally?
- A. zebra
 - B. tortoise
 - C. pelican
 - D. blue whale
23. Omnivores eat both plant and animal foods. Which animal below is an omnivore?
- A. skunk
 - B. polar bear
 - C. wolf
 - D. deer
24. Which of the following is part of the human digestive system?
- A. artery
 - B. spinal cord
 - C. bronchial tube
 - D. small intestine
25. Its common name is lockjaw. Which disease below causes a person's muscles to contract, making it hard to open the mouth or swallow?
- A. diphtheria
 - B. rabies
 - C. tetanus
 - D. pertussis
26. The diagram here shows the parts of what piece of lab equipment?
- A. graduated cylinder
 - B. Bunsen burner
 - C. funnel
 - D. test tube
27. Which choice below names a scientist who studies the atmosphere?
- A. meteorologist
 - B. botanist
 - C. zoologist
 - D. entomologist
28. As part of the scientific method, a scientist predicts the experiment's outcome based on observations and other information. Which of the following words names that prediction?
- A. law
 - B. estimate
 - C. hypothesis
 - D. theory
29. He co-founded the American Telephone and Telegraph Company (now AT&T) in 1885. Which man below is credited with inventing and patenting the first practical telephone?
- A. Edward Jenner
 - B. George Washington Carver
 - C. Alexander Graham Bell
 - D. Jonas Salk



30. Match each scientist with her field(s).

- x. Florence Nightingale
- y. Rachel Carson
- z. Marie Curie

- 1. medicine
- 2. physics and chemistry
- 3. ecology

- A. x-2, y-3, z-1
- B. x-1, y-2, z-3
- C. x-3, y-2, z-1
- D. x-1, y-3, z-2

TIEBREAKERS 31-36

31. It sits at the end of the Little Dipper's handle. Which choice below names Earth's current North Star?
- A. Vega
 - B. Polaris
 - C. Rigel
 - D. Sirius
32. Which of the following is NOT a characteristic of monocots?
- A. pollen with one furrow or pore
 - B. flower parts in multiples of 3
 - C. parallel leaf veins
 - D. embryo with 2 cotyledons
33. Certain invertebrate species like the cockroach and grasshopper undergo incomplete metamorphosis. Which choice below names that process's middle stage?
- A. nymph
 - B. larva
 - C. scarab
 - D. pupa
34. Three of the following animals are placental mammals. Which one is a marsupial?
- A. horse
 - B. rhinoceros
 - C. kangaroo
 - D. bison
35. Below are four parts of an electrical circuit. Which one is what the electricity is powering, such as a light bulb?
- A. conductor
 - B. load
 - C. power source
 - D. switch
36. On July 21, 1969, which of the following men became the first person to set foot on the moon?
- A. Neil Armstrong
 - B. Louis Pasteur
 - C. Galileo Galilei
 - D. Alfred Nobel

**Science – Elementary – District
2023 Governor’s Cup**

1. C
2. B
3. A
4. D
5. D
6. B
7. C
8. A
9. D
10. A
11. D
12. C
13. A
14. B
15. D
16. A
17. C
18. B
19. B

20. C
21. D
22. B
23. A
24. D
25. C
26. B
27. A
28. C
29. C
30. D
31. B
32. D
33. A
34. C
35. B
36. A

TIEBREAKERS 31-36